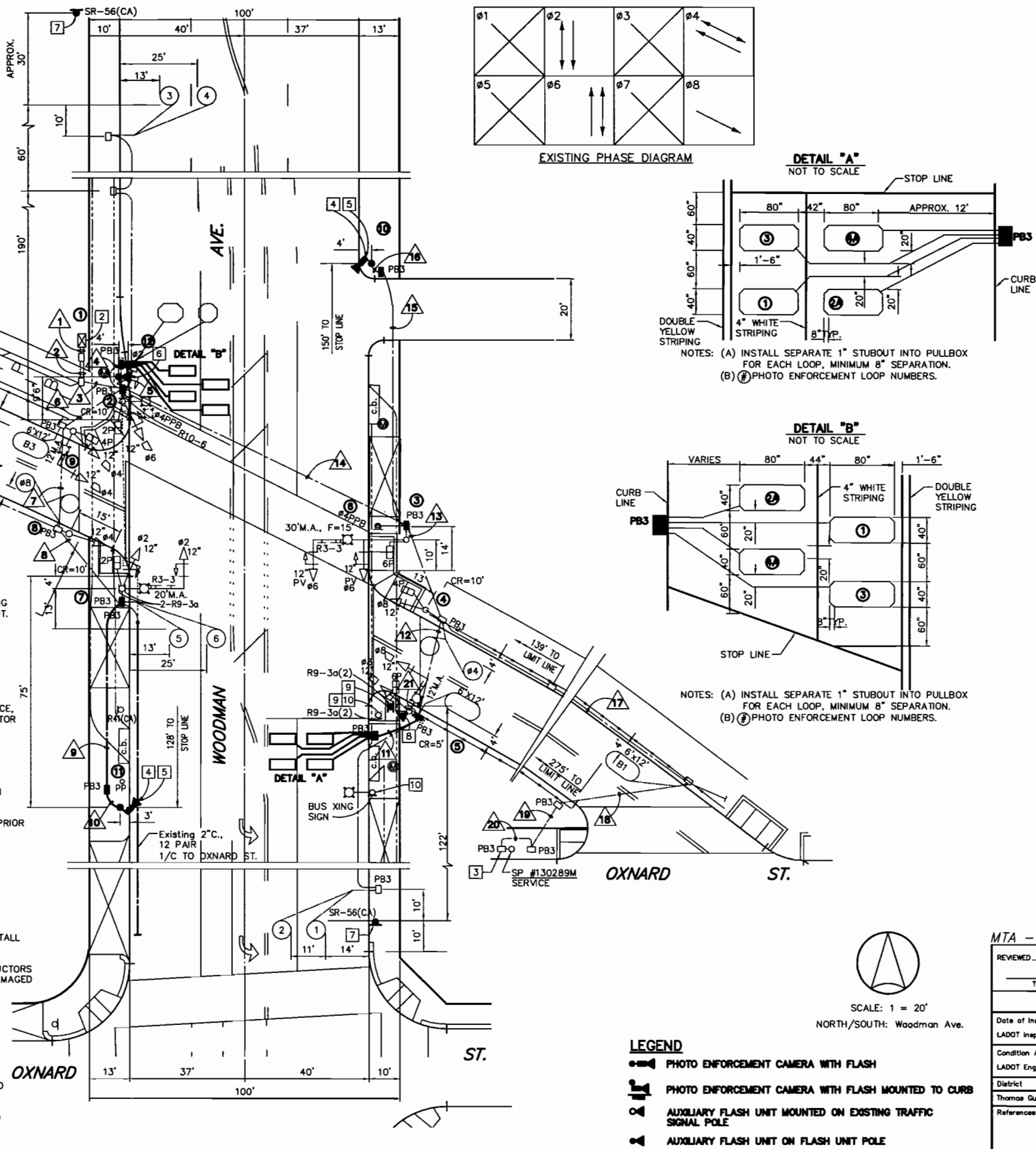


SIGNAL STANDARD SCHEDULE					
No.	TYPE	FOUND. TYPE	PED. PUSH BUTN TYPE	PHASE	REMARKS
1	332 CAB. (E)	F-332			ATSAC 2070 CONTROLLER IN 332 CAB.
2	CD9530* (E)	*	6"	4	
3	19-3-80 (E)	CIDH**			30' M.A., F=15'
4	1 (E)	F-1			CUSTOM 12' M.A.
5	CD954 (E)	F-2			
6	7 (E)	***	PTM	4	20' M.A.
7	CD954 (E)	F-2			CUSTOM 12' M.A.
8	1 (E)	F-1			
9	CD954 (E)	F-2			
10	CAMERA POLE	****			PHOTO ENFORCEMENT CAMERA POLE
11	CAMERA POLE	****			PHOTO ENFORCEMENT CAMERA POLE
12	FLASH UNIT POLE	****			PHOTO ENFORCEMENT FLASH UNIT POLE

(E) Existing
 * For reference only; see separate Street Lighting sheet
 ** CIDH PILE Foundation as per Caltrans Latest Standard Plans.
 *** Alternate F-7 Foundation
 **** See Drawing No. N-001B Detail "E"
 ***** See Drawing No. N-002 Detail "C"

NOTES:

- SEMI-ACTUATED OPERATION WITH TRANSIT PRIORITY AND RED LIGHT PHOTO ENFORCEMENT (ATSAC 2070 CONTROLLER IN 332 CABINET)
 #2 - WOODMAN AVE. SB & W/S PED XING
 #4 - BUSWAY WB & N/S PED XING
 #6 - WOODMAN AVE. NB & E/S PED XING
 #8 - BUSWAY EB VICTORY CORRIDOR EAST ATSAC SYSTEM DIRECT WIRE (CABLE) INTERCONNECT
- CONTRACTOR TO INSTALL TWO PANELS FURNISHED BY ACS, EACH PANEL WITH 4 CURRENT SENSING SWITCHES (CR MAGNETICS CR9350-ACA), ON CONTROLLER CABINET FRAME AS DIRECTED BY LADOT. INSTALL IN-LINE WATERPROOF FUSE HOLDER WITH 5A FUSE IN SWITCH OUTPUT TO CAMERA UNIT.
- CONTRACTOR TO INSTALL WATERPROOF 120VAC SERVICE CONNECTORS FOR EACH PHOTO ENFORCEMENT CAMERA AT SERVICE PULLBOX. PROVIDE IN-LINE WATERPROOF FUSE HOLDER WITH 15A FUSE IN EACH CAMERA SERVICE CIRCUIT. SERVICE FROM SP #130289M LOCATED ON NW CORNER OF THE INTERSECTION OF BUFFALO AVE., OXNARD ST. AND BUSWAY.
- CONTRACTOR TO INSTALL PHOTO ENFORCEMENT CAMERA POLE PER DWG. NO. N-001B DETAIL "D", FURNISHED BY ACS, ON FOUNDATION PER DWG. NO. N-001B DETAIL "E". CONNECT CAMERA SERVICE, BRT RED AND YELLOW SENSE WRES, AUXILIARY FLASH POWER AND TRIGGER CONDUCTORS, DETECTOR LEAD-IN CABLES, AND GROUND AT CAMERA POLE TERMINAL BLOCK.
- CONTRACTOR TO INSTALL GROUND ROD AND CONNECT 10AWG SOLID GROUND WIRE TO PHOTO ENFORCEMENT CAMERA POLE TERMINAL BLOCK GROUND LUG.
- CONTRACTOR TO INSTALL AUXILIARY FLASH UNIT AND FLASH UNIT POLE PER DWG. NO. N-002 DETAIL "C", FURNISHED BY ACS. MAKE CONNECTIONS AT FLASH UNIT TERMINAL BLOCK FOR FLASH UNIT POWER AND TRIGGER CONDUCTORS.
- CONTRACTOR TO INSTALL ACS-FURNISHED PHOTO ENFORCEMENT WARNING SIGN AND SIGN POLE, PRIOR TO ACTIVATION OF SYSTEM.
- CONTRACTOR TO INSTALL AUXILIARY FLASH UNIT, FURNISHED BY ACS, ON EXISTING LADOT TRAFFIC SIGNAL POLE AT MINIMUM 10'-0" ABOVE SIDEWALK GRADE. ADJUST EXISTING SIGNS ATTACHED TO TRAFFIC SIGNAL POLE, AS NEEDED, TO ALLOW FOR FLASH UNIT INSTALLATION. MAKE CONNECTION AT FLASH UNIT TERMINAL BLOCK FOR FLASH UNIT POWER AND TRIGGER CONDUCTORS.
- CONTRACTOR TO RELOCATE EXISTING TRAFFIC SIGN POLE AND TWO R9-3a SIGNS AS INDICATED.
- CONTRACTOR TO REMOVE EXISTING "BUS XING" SIGN FROM EXISTING TRAFFIC SIGN POLE, AND INSTALL ON STREET LIGHTING POLE AS INDICATED.
- CONTRACTOR TO PROTECT IN PLACE EXISTING TRAFFIC SIGNAL PULLBOXES, CONDUITS, AND CONDUCTORS DURING CONSTRUCTION. REPLACE ANY PULLBOX LIDS, PULLBOXES, CONDUITS, OR CONDUCTORS DAMAGED DURING CONSTRUCTION AT CONTRACTOR'S SOLE EXPENSE.
- PHOTO ENFORCEMENT LOOPS SHALL BE 3 TURNS IMSA 51-7 14AWG WIRE IN MAXIMUM 2" DEEP, 1/4-INCH WIDE SAWCUT. LOOP LEAD-IN WRES TO BE TWISTED MINIMUM FOUR TURNS PER FOOT.
- CONTRACTOR TO SPLICE LOOP WRES TO DLC AT PULLBOX BY CRIMPING THEN SOLDERING COMPRESSION FITTING TO CONDUCTOR ENDS, THEN ENCASE IN COMPRESSION FITTING AND CONDUCTOR ENDS WITH HEAT SHRINK TUBING AND VINYL ELECTRICAL TAPE.
- CONTRACTOR TO CONTACT RAY MAY AT (619) 247-1252 FOR ACS-FURNISHED CAMERA POLE AND AUXILIARY FLASH UNIT INSTALLATION AND CONDUCTOR TERMINATION DETAILS.
- CONTRACTOR TO CONTACT TODD REDMOND AT (213) 439-6219 FOR PICK-UP OF ACS-FURNISHED EQUIPMENT FIVE (5) WORKING DAYS PRIOR TO PICK-UP DATE.



CONDUCTOR SCHEDULE												
SIZE No.	CABLE / WIRE	RUN										
3C12	PHOTO ENFORCEMENT SERVICE	1	1	1	1	1	1	1	1	1	1	1
5C14	PHOTO ENFORCEMENT ENFORCED RED AND YELLOW	2	2	2	2	2	2	2	2	2	2	2
2C12	PHOTO ENFORCEMENT DLC	4	4	4	4	4	4	4	4	4	4	4
5C14	PHOTO ENFORCEMENT AUXILIARY FLASH	1	1	1	1	1	1	1	1	1	1	1
CONDUIT SIZE		2"	2"	2"	2"	2"	2"	2"	2"	2"	2"	2"
NEW/EXISTING		E	E	E	E	E	E	E	E	E	E	E

(*) IMSA 50-2
 (**) IMSA 40-2

NOTICE TO CONTRACTOR

- EXCEPT WHERE OTHERWISE INDICATED, MATERIALS AND INSTALLATION METHODS SHALL CONFORM TO THE LATEST EDITION, WITH SUPPLEMENTS, OF THE STANDARD SPECIFICATION FOR PUBLIC WORKS CONSTRUCTION, ADOPTED BY THE CITY OF LOS ANGELES BOARD OF PUBLIC WORKS (LAPBW), INCLUDING THE LATEST VERSION OF STANDARD PLAN S-610 ISSUED BY THE LAPBW, AND THE LATEST EDITION, WITH AMENDMENTS, OF THE SPECIAL PROVISIONS AND STANDARD DRAWINGS FOR THE INSTALLATION AND MODIFICATION OF TRAFFIC SIGNALS, ISSUED BY THE CITY OF LOS ANGELES DEPARTMENT OF TRANSPORTATION (LADOT).
- THE LADOT TRAFFIC SIGNAL INSPECTOR SHALL BE NOTIFIED AT (213) 792-7637. THE SENIOR MAINTENANCE SUPERVISOR AT (818) 756-7846, AND THE LADOT MAJOR CONSTRUCTION COORD. DIVISION (MCCD) AT (213) 485-2610 THREE (3) WORKING DAYS PRIOR TO THE BEGINNING OF CONSTRUCTION.
- NORMAL TRAFFIC CONTROL SHALL BE MAINTAINED FROM 6:00 A.M. TO 9:00 A.M. AND 3:00 P.M. TO 7:00 P.M. ON WEEKDAYS. HOURS AND DURATION WHEN TRAFFIC SIGNALS ARE DEACTIVATED AND/OR PLACED INTO FLASHING OPERATION REQUIRE THE PRIOR APPROVAL OF THE LADOT METRO RAIL ENGINEER. TRAFFIC SIGNAL DEACTIVATION AND/OR FLASHING OPERATION SHALL BE IMPLEMENTED BY A LADOT TRAFFIC SIGNAL ELECTRICIAN, IN COORDINATION WITH THE ATSAC CONTROL CENTER, AND SHALL BE REQUESTED THREE (3) WORKING DAYS IN ADVANCE THROUGH THE LADOT MCCD, (213) 485-2610. FULL RESTORATION OF THE ATSAC SYSTEM MUST OCCUR WITHIN FIVE (5) WORKING DAYS AFTER DEACTIVATION.
- THE CONTRACTOR SHALL COORDINATE WITH THE LADOT SIGNAL SHOP AT (213) 847-2944 TO ARRANGE CONTROLLER DELIVERY. THE CONTRACTOR SHALL PROVIDE AND DELIVER A COMPLETE FUNCTIONING CONTROLLER ASSEMBLY AND CABINET TO THE LADOT SIGNAL SHOP THIRTY (30) CALENDAR DAYS PRIOR TO THE INSTALLATION DATE, AND PICK UP THE CONTROLLER WITHIN FIFTEEN (15) CALENDAR DAYS AFTER NOTIFICATION BY LADOT SIGNAL SHOP.
- ALL REMOVED TRAFFIC SIGNAL EQUIPMENT IDENTIFIED AS SALVAGED MATERIAL SHALL BE RETURNED TO LADOT. ARRANGEMENTS FOR RETURNING IDENTIFIED EQUIPMENT TO LADOT SHALL BE MADE FIVE (5) WORKING DAYS IN ADVANCE BY CONTACTING THE LADOT MATERIAL CONTROL DIVISION AT (213) 928-9638. THE CONTRACTOR SHALL OBTAIN A SALVAGE RECEIPT FROM LADOT AND FORWARD COPY, SIGNED BY THE CONTRACTOR AND THE RECEIVING LADOT YARD TO LADOT MCCD AT 555 RAMIREZ ST., SUITE 315, L.A. CA 90012. ALL OTHER REMOVED SIGNAL EQUIPMENT SHALL BE RETAINED BY THE CONTRACTOR.
- ALL EXISTING TRAFFIC CONTROL SIGNING SHALL BE MAINTAINED WITH STANDARD SIGN BRACKETS BY THE CONTRACTOR. EXCEPT WHERE OTHERWISE INDICATED, EXISTING LARGE STREET NAME SIGNS ON MAST ARM TYPE STANDARDS SHALL BE RELOCATED FROM THE VERTICAL POLE TO THE MAST ARM PER LADOT STANDARD DRAWING NO. S-457.0. WHEN CALLED OUT ON THIS PLAN, CONTRACTOR SHALL FURNISH AND INSTALL LARGE STREET NAME SIGNS AND SIGN BRACKETS FOR INSTALLATION ON MAST ARM PER LADOT STD. DWG. NO. S-457.0. ANY TRAFFIC SIGNING DAMAGED OR LOST BY THE CONTRACTOR SHALL BE REPLACED AT THE CONTRACTOR'S EXPENSE.
- THE CONTRACTOR SHALL FURNISH AND INSTALL, AS SHOWN ON THE PLANS, ALL TRAFFIC SIGNAL EQUIPMENT SUCH AS STANDARDS, MAST ARMS, SIGNAL HEADS, AND OTHER EQUIPMENT AS REQUIRED. ALL SUCH EQUIPMENT SHALL BE NEW. ALL SUCH EQUIPMENT SHALL BE IN ACCORDANCE WITH THE LATEST EDITION OF THE SIGNAL MANUAL. ALL VEHICLE HEAD EQUIPMENT SUPPLIED BY THE CONTRACTOR SHALL HAVE 1-1/2 INCH DIAMETER PIPE FIT MOUNTING HARDWARE, INCLUDING ALL TOP MOUNT HEADS. PED HEAD MODULE REPLACEMENTS SHOWN ON THE PLAN SHALL REQUIRE A COMPLETE INSTALLATION, INCLUDING NEW PED HEAD AND TERMINAL COMPARTMENT. LEFT TURN SIGNAL HEADS ON MAST ARMS WITH TWO OR MORE SIGNAL HEADS SHALL BE TOP MOUNTED.

THIS PLAN SUPERSEDES PLAN
 SIGNED ON APRIL 17, 2006

MTA - SAN FERNANDO VALLEY RAPID BUS SYSTEM (PHOTO ENFORCEMENT)

REVIEWED _____ 20 _____	RECOMMENDED _____ 20 _____	ACCEPTED _____
Transportation Engineer	Senior Transportation Engineer	Principal Transportation Engineer
CITY OF LOS ANGELES DEPARTMENT OF TRANSPORTATION GLORIA JEFF, GENERAL MANAGER		
BUSWAY AND WOODMAN AVE.		
Date of Installation: _____	TITLE	
LADOT Inspector: _____	District: East Valley	
Condition As Of: _____	Thomas Guide: 532-C7	
LADOT Engineer: _____	References	
File Name: 33222-R1.DWG	Intersection No: 33222	Project No: 74720
NOTICE: IF PLAN HAS NOT BEEN IMPLEMENTED WITHIN TWO (2) YEARS OF THE ACCEPTANCE DATE, IT MUST BE RESUBMITTED TO LADOT SIGNAL DESIGN SECTION FOR REVIEW AND APPROVAL.		

DESIGNED BY: J. CURRY
 DRAWN BY: D. BARRAZA
 CHECKED BY: K. KIM
 IN CHARGE: J. CURRY
 DATE: _____

LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
 SAN FERNANDO VALLEY EAST-WEST BRT PROJECT

ACS GOVERNMENT SOLUTIONS

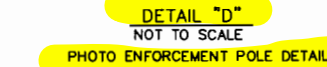
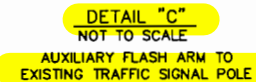
PLANS PREPARED BY: PB FARRADYNE
 1000 S. GILBERT STREET
 SUITE 300
 SAN ANTONIO, TEXAS 78204
 TEL: (214) 343-3333
 FAX: (214) 343-3333

REV	DATE	BY	APP	REG NO	EXPIRES	SEAL HOLDER	DESCRIPTION

METRO ORANGE LINE PHOTO
 ENFORCEMENT SYSTEM
 INSTALLATION PLAN
 BUSWAY & WOODMAN AVE.

SEGMENT NO	
DRAWING NO	N-003
SCALE	1" = 20'
SHEET NO	

* For reference only; see separate Street Lighting sheet
 ** See Detail "E"
 *** CIDH PILE Foundation as per LADOT Std. Dwg. No. S-52.1-4B.



(*) IMSA 50-2
(**) IMSA 4D-2

NOTES:

1. SEMI-ACTUATED OPERATION (ATSC 2070 CONTROLLER WITH TRANSIT PRIORITY IN 332 CABINET)
 - #2 - TUJUNGA AVE. SB & W/S PED. XING (AT CHANDLER BLVD NORTH RDWY)
 - #3 - CHANDLER BLVD (NORTH RDWY) WB, WB LT & N/S PED XING (EXCLUSIVE PHASE)
 - #4 - TUJUNGA AVE ES PED XING (AT CHANDLER BLVD NORTH RDWY)
 - #6 - TUJUNGA AVE NB & E/S PED XING (AT CHANDLER BLVD SOUTH RDWY)
 - #7 - CHANDLER BLVD (SOUTH RDWY) EB & S/S PED XING
 - #8 - CHANDLER BLVD (SOUTH RDWY) WB RT, WB LT & WS PED XING
 - OLA - (#2 & #3) TUJUNGA AVE SB AT BRT
 - OLB - (#2 & #3) TUJUNGA AVE SB (AT CHANDLER BLVD SOUTH RDWY)
 - OLC - (#4 & #6) TUJUNGA AVE NB AT BRT
 - OLD - (#4 & #6) TUJUNGA AVE NB (AT CHANDLER BLVD NORTH RDWY)
 - OLE - (#7) CHANDLER BLVD (SOUTH RDWY) EB LT
 - OLF - (#3 #4 & #6) TUJUNGA AVE E/S & W/S PED XING AT BRT
 - BRT A - BUSWAY WB
 - BRT B - BUSWAY EB
- DIRECT WIRE (CABLE) INTERCONNECT
VENTURA 2B ATSC SYSTEM
2. CONTRACTOR TO INSTALL PHOTO ENFORCEMENT CAMERA POLE, FURNISHED BY ACS, ON FOUNDATION PER DETAIL "E". CONNECT CAMERA SERVICE, BRT RED AND YELLOW SENSE WIRES, AUXILIARY FLASH POWER AND TRIGGER CONDUCTORS, DETECTOR LEAD-IN CABLES, AND GROUND AT CAMERA POLE TERMINAL BLOCK.
3. CONTRACTOR TO INSTALL GROUND ROD AND CONNECT 10AWG SOLID GROUND WIRE TO PHOTO ENFORCEMENT CAMERA POLE TERMINAL BLOCK GROUND LUG.
4. CONTRACTOR TO FURNISH AND INSTALL FLASH UNIT MOUNTING ARM ON EXISTING LADOT TYPE 1A TRAFFIC SIGNAL POLE AT MINIMUM 10'-0" ABOVE SIDEWALK GRADE, AND INSTALL AUXILIARY FLASH UNIT FURNISHED BY ACS. ADJUST EXISTING SIGNS ATTACHED TO TRAFFIC SIGNAL POLE, AS NEEDED, TO ALLOW FOR FLASH UNIT INSTALLATION. MAKE CONNECTIONS AT FLASH UNIT TERMINAL BLOCK FOR FLASH UNIT POWER AND TRIGGER CONDUCTORS.
5. CONTRACTOR TO INSTALL ACS-FURNISHED PHOTO ENFORCEMENT WARNING SIGN AND SIGN POLE, PRIOR TO ACTIVATION OF SYSTEM.
6. CONTRACTOR TO PROTECT IN PLACE EXISTING TRAFFIC SIGNAL PULLBOXES, CONDUITS, AND CONDUCTORS DURING CONSTRUCTION. REPLACE ANY PULLBOX LIDS, PULLBOXES, CONDUITS, OR CONDUCTORS DAMAGED DURING CONSTRUCTION AT CONTRACTOR'S SOLE EXPENSE.
7. PHOTO ENFORCEMENT LOOPS SHALL BE 3 TURNS IMSA 51-7 14AWG WIRE IN MAXIMUM 2" DEEP, 1/4-INCH WIDE SAWCUT. LOOP LEAD-IN WIRES TO BE TWISTED MINIMUM FOUR TURNS PER FOOT.
8. CONTRACTOR TO SPLICE LOOP WIRES TO DLC AT PULLBOX BY CRIMPING THEN SOLDERING COMPRESSION FITTING TO CONDUCTOR ENDS, THEN ENCASING COMPRESSION FITTING AND CONDUCTOR ENDS WITH HEAT SHRINK TUBING AND VINYL ELECTRICAL TAPE.
9. CONTRACTOR TO CONTACT RAY MAY AT (619) 247-1252 FOR ACS-FURNISHED CAMERA POLE AND AUXILIARY FLASH UNIT INSTALLATION AND CONDUCTOR TERMINATION DETAILS.
10. CONTRACTOR TO CONTACT TODD REDMOND AT (213) 439-6219 FOR PICK-UP OF ACS-FURNISHED EQUIPMENT FIVE (5) WORKING DAYS PRIOR TO PICK-UP DATE.

THIS PLAN SUPERSEDES PLAN
SIGNED ON MARCH 19, 2006

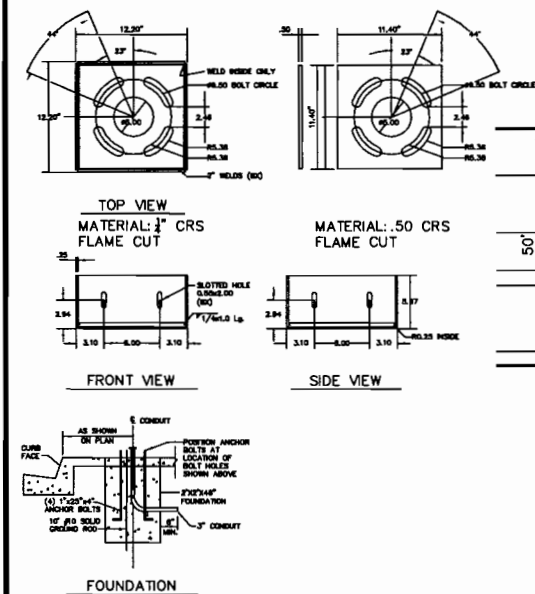
MTA - SAN FERNANDO VALLEY RAPID BUS SYSTEM (PHOTO ENFORCEMENT)

REVIEWED _____ 20 _____ _____ Transportation Engineer	RECOMMENDED _____ 20 _____ _____ Senior Transportation Engineer	ACCEPTED _____ _____ Principal Transportation Engineer
_____ Date of Installation: _____ LATOT Inspector: _____ _____ Condition As Of: _____ LATOT Engineer: _____ _____ District EAST VALLEY Thomas Guide 562-J2	CITY OF LOS ANGELES DEPARTMENT OF TRANSPORTATION GLORIA JEFF, GENERAL MANAGER TITLE CHANDLER BLVD., TUJUNGA AVE. & BUSWAY	
References	File Name 32810-R2.DWG	Intersection No. 32810
		Project No. 74720

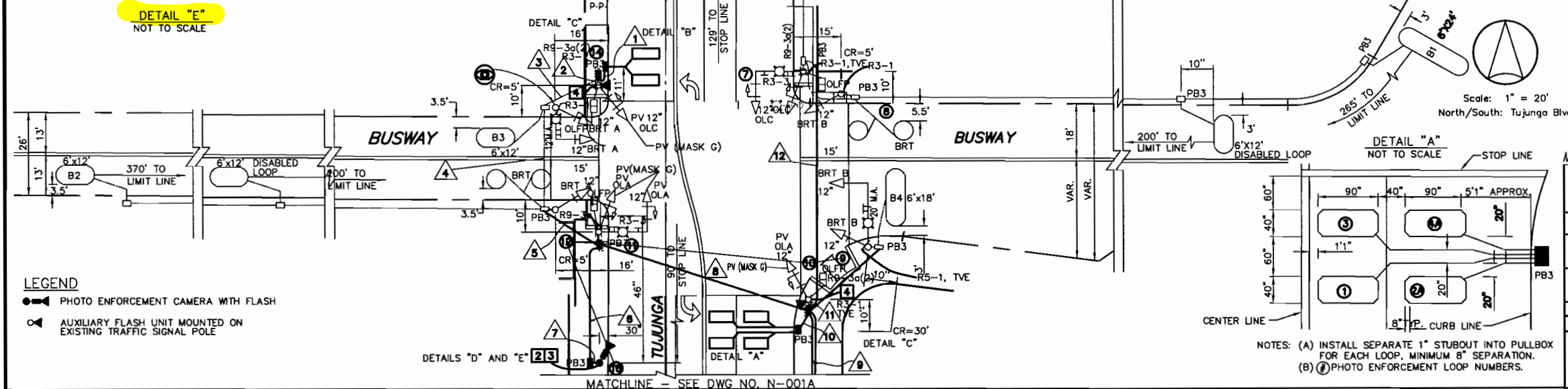
NOTICE: IF PLAN HAS NOT BEEN IMPLEMENTED WITHIN TWO (2) YEARS OF THE ACCEPTANCE DATE, IT MUST BE RESUBMITTED TO LAROT. SPECIAL SPECIAL FOR REVIEW AND ACTION.

METRO ORANGE LINE PHOTO
ENFORCEMENT SYSTEM
INSTALLATION PLAN
CHANDLER BLVD., TUJUNGA AVE.
AND BUSWAY

SEGMENT NO	
DRAWING NO N-001B	REV
SCALE 1" = 20'	
SHEET NO	



DETAIL "E"
NOT TO SCALE

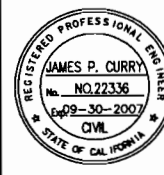


LEGEND

- PHOTO ENFORCEMENT CAMERA WITH FLASH
 ◐ AUXILIARY FLASH UNIT MOUNTED ON EXISTING TRAFFIC SIGNAL POLE

[illegible]

DESIGNED BY	K. KIM
DRAWN BY	D. BARRAZA
CHECKED BY	J. CURRY
IN CHARGE	J. CURRY
DATE	


LOS ANGELES COUNTY METROPOLITAN TRANSPORTATION AUTHORITY
SAN FERNANDO VALLEY EAST-WEST BRT PROJECT

REVIEWED BY



PLANS PREPARED BY
PB FARRADYNE
444 S. FLOWER STREET
SUITE 3700
LOS ANGELES, CA 90071
PH: (213) 362-8470
FAX: (213) 362-9480

JAMES P. CLURRY 22336
REG. CIVIL ENGINEER NO.

JAMES P. CURRY	22334	08/30/70
REG. CIVIL ENGINEER	NO.	DATE